

US-SPEC-2024-9102982-A1

## Microtonal Mineral Lithophone with Electromagnetically Driven Solenoid Mallets (Series D)

Filing: 2024-10-22 • Status: DEFENSIVE PUBLICATION • Discipline: Applied Anomalies  
Inventors: Elena Mstislav • Assignee: Zazie Institute of Applied Anomalies / Zazie Productions LLC  
Canonical: <https://zazieinstitute.org/patents/pat-2024-022>

### ABSTRACT & METHOD DISCLOSURE

An acoustic instrument comprising tuned mineral tone slabs subjected to precision mechanical strikes by solenoid actuators, capturing non-harmonic modal decay via high-impedance piezoelectric pickups.

### INDEPENDENT CLAIMS

1. An apparatus for experimental sound transduction, comprising: an acoustic input transducer configured to convert physical vibrations into an analog electrical signal; a dynamic hysteresis conditioning stage; and an acoustic output transducer configured to project conditioned sound into an architectural boundary.
2. The apparatus of claim 1, further comprising a real-time digital microcontroller configured to monitor spectral peaks and adjust closed-loop phase alignment.
3. A method for algorithmic spatial sound diffusion, comprising: sampling multichannel audio streams; calculating distance-based amplitude weights across distributed speaker coordinates; and routing low-latency audio packets over an asynchronous IP network.

### DEPENDENT CLAIMS

4. The apparatus of claim 1, wherein the dynamic conditioning stage comprises a soft-clipping JFET transistor network.
5. The apparatus of claim 1, wherein the acoustic output transducer comprises an exponential horn with aluminum phase plugs.
6. The method of claim 3, wherein the network protocol is Dante operating under IEEE 1588 precision time protocol.
7. The apparatus of claim 1, wherein the input transducer comprises a dual-axis optical laser triangulation sensor.

### PRIOR-ART CRITIQUE

Contrasted with traditional commercial DAW plugins and static acoustic room treatments, the present speculative invention couples physical hardware dynamics directly to room boundaries, enabling adaptive generative performance without artificial digital latency.

### COUNSEL MEMO

*Defensive publication strategy approved under ZIAA Open Research Charter. Discloses all core mathematical and physical topologies to prevent predatory patent encumbrance by proprietary audio conglomerates.*